

12(a)	1 not affected by external factors	B1
	2 cannot predict when a (particular) nucleus will decay or cannot predict which nucleus will decay (next)	B1
12(b)(i)	Number of atoms = $\frac{1.0 \times 10^{-9}}{90 \times 1.66 \times 10^{-27}}$ or $\frac{1.0 \times 10^{-9} \times 6.02 \times 10^{23}}{90 \times 10^{-3}}$ $= 6.693 \times 10^{15}$	C1
	$A = \lambda N$ $\lambda = \frac{5.2 \times 10^6}{6.693 \times 10^{15}}$	C1
	$\lambda = 7.8 \times 10^{-10} \text{ s}^{-1}$	A1
12(b)(ii)	daughter nucleus is unstable	B1